JAKOB M. HELTON

jakobhelton@psu.edu ◇ +1 (304) 360 0337
jakobhelton.github.io ◇ orcid.org/0000-0003-4337-6211
Department of Astronomy and Astrophysics, The Pennsylvania State University

SUMMARY & SCHOLARLY IDENTITY

Funding – roughly **$950,000** in grants/awards secured over the past **2** years as PI

Publications – **8** first author, **5** second author, **104** in total (**10,595** citations, $\boldsymbol{h = 46}$)

Research Interests – galaxy formation and evolution; high-redshift galaxies and galaxy clusters; the large-scale structure of the Universe; observational extragalactic astronomy

Professional Expertise – infrared imaging and spectroscopy with JWST (designing and optimizing observing strategies, developing data reduction pipelines, and implementing analysis techniques); panchromatic spectral energy distribution (SED) fitting

Collaborations – JWST/NIRCam Instrument Science Team; JWST/MIRI Instrument Science Team; the JWST Advanced Deep Extragalactic Survey (JADES) as the working group lead for grism science, including the large-scale environment

Observing – PI/Co-I of numerous programs working with NASA flagship space missions (Roman, JWST) and ground-based observatories (ALMA, Keck, Magellan, MMT)

EMPLOYMENT

The Pennsylvania State University *August 2025 - Present*
Position – Evolving Universe Postdoctoral Fellow *[Mentor – Joel Leja]*

EDUCATION

The University of Arizona *August 2021 - July 2025*
Degree – Ph.D. and M.S. in Astronomy and Astrophysics
Thesis – *At the Break of Cosmic Dawn: Identifying and Understanding the Most Distant Galaxies and Galaxy Clusters with JWST* *[Advisors – Marcia Rieke; Kevin Hainline]*

Princeton University *September 2017 - May 2021*
Degree – A.B. in Astrophysical Sciences with High Honors
Thesis – *The Nebular Properties of Star-Forming Galaxies at Intermediate Redshifts from the Large Early Galaxy Astrophysics Census* *[Advisors – Allison Strom; Jenny Greene]*

PUBLICATIONS

Link to SciX Public Library *Last Updated: 3 November 2026*

Summary of All Publications number: 104; citations: 10,595; h-index: 46
Summary of Refereed Publications number: 73; citations: 10,176; h-index: 44

List of All Publications *Continues Later, Starting on Page 5*

TELESCOPE ALLOCATIONS

Observing Programs as Principal Investigator (PI) or Co-PI

*Nancy Grace Roman Space Telescope (**1** program, archival/analysis, with $**450,000** in funding received):* Cycle 1 Analysis #19100

*James Webb Space Telescope (**3** programs, **117.2** hours, $**499,627** in funding received):* Cycle 5 GO #11376; Cycle 4 GO #08544; Cycle 3 GTO #04549

*MMT Telescope (**1** program, **2.0** nights using Binospec)*

Observing Programs as Co-Investigator (Co-I)

*Nancy Grace Roman Space Telescope (**2** programs, archival/analysis, with $**300,000** in funding received)*: Cycle 1 Analysis #19082; Cycle 1 Analysis #19067

*James Webb Space Telescope (**14** programs, **590.3** hours of primary telescope time and **1000.0** hours of pure parallel time, with more than $**2,500,000** in funding received):* Cycle 5 GO #12396; Cycle 5 GO #10790; Cycle 5 GO #10518; Cycle 5 GO #10445; Cycle 5 GO #10311; Cycle 4 GO #08018; Cycle 4 GO #07935; Cycle 4 GO #07335; Cycle 2 GO #03659; Cycle 2 GO #03577; Cycle 2 GO #03215; Cycle 2 GO #02959; Cycle 2 GO #02883; Cycle 3 GTO #04540

*Atacama Large Millimeter/sub-millimeter Array (**2** programs, **33.4** hours of primary telescope time):* Cycle 10 #2023.1.00391.S; Cycle 10 #2023.1.00336.S

*Magellan Telescopes (**2** programs, **11.5** nights using FIRE and IMACS)*

*Keck Telescopes (**1** program, **0.5** nights using MOSFIRE)*

*MMT Telescope (**1** program, **3.0** nights using Binospec)*

STUDENTS SUPERVISED

Graduate Students

Marta Laska (The Pennsylvania State University, primary supervisor)

Undergraduate Students

Natasha Patel (The Pennsylvania State University, supervising honors thesis)
Tristen Shields (The University of Arizona, now a graduate student at Yale University)

SERVICE & LEADERSHIP

Telescope Allocation Committees – *Subaru Telescope; ALMA (distributed review)*

Invited Peer Reviewer – *The Astrophysical Journal (ApJ); Monthly Notices of the Royal Astronomical Society (MNRAS); Astronomy & Astrophysics; Astronomy & Computing*

Member – *Climate and Diversity Committee (PSU Astronomy)* *Oct. 2026 - Present*
Member – *Graduate Program Committee (PSU Astronomy)* *Sep. 2026 - Present*
Mentor – *Mentoring Program for ECRs (PENGUIN)* *Aug. 2026 - Present*

Leader – *Working Group, Grism Science (JADES Collaboration)* *Aug. 2024 - Present*
Organizer – *UofA's Extragalactic Seminar (EURECA)* *Jun. 2023 - Jun. 2025*
Mentor – *Summer Research Program (Carnegie Obs.)* *Jun. 2020 - Jun. 2025*
Organizer – *Princeton's Galaxy Seminar (Galread)* *Jan. 2020 - Aug. 2020*

TEACHING EXPERIENCE

ATOMM, Astronomy Tutoring For Majors *(Tutor, UofA)* *Jan. 2025 - May 2025*

ASTR333, Astronomy and the Arts *(Teaching Assistant, UofA)* *Jan. 2024 - May 2024*

PHY103, Physics I, Mechanics *(Teaching Assistant, Princeton)* *Sep. 2019 - Jan. 2020*
PHY104, Physics II, E&M *(Teaching Assistant, Princeton)* *Jan. 2019 - May 2019*

MAT202: Linear Algebra *(Head Tutor, Princeton)* *Jan. 2018 - May 2021*
MAT201: Multi-Variable Calculus *(Head Tutor, Princeton)* *Jan. 2018 - May 2021*
MAT104: Single-Variable Calculus *(Head Tutor, Princeton)* *Jan. 2018 - May 2021*
MAT103: Single-Variable Calculus *(Head Tutor, Princeton)* *Jan. 2018 - May 2021*

SELECTED MEDIA COVERAGE

The University of Arizona | 13 March 2025
James Webb Space Telescope reveals unexpected complex chemistry in primordial galaxy

The University of Arizona | 3 October 2024
Eyes on the Universe | How UofA Researchers Confirmed Farthest Galaxy with JWST

Space.com | 15 July 2024
Can the James Webb Space Telescope see galaxies over the universe's horizon?

Astronomy Magazine | 6 June 2024
Webb discovers the earliest known galaxy – for now

Big Think | 3 June 2024
5 big lessons from JWST's new record-setting galaxy

Space.com | 30 May 2024
James Webb Space Telescope spots the most distant galaxy ever seen

NASA Webb Mission Team | 30 May 2024
NASA's James Webb Space Telescope Finds Most Distant Known Galaxy

The University of Arizona | 30 May 2024
Webb Telescope spots the two most distant galaxies ever seen at cosmic dawn

University of Cambridge | 30 May 2024
Earliest, most distant galaxy discovered with James Webb Space Telescope

JADES | 30 May 2024
Earliest, most distant galaxy discovered with James Webb Space Telescope Galaxy dates back to 300 million years after the Big Bang

SELECTED PRESENTATIONS

† = invited presentation

JWST Conference, "Towards the Full Bloom"	*Tokyo, Japan; Mar. 2026*
† University of Pittsburgh's AstroLunch Seminar	*Pittsburgh, PA; Mar. 2026*
† Monday Afternoon Talk at MIT's Kavli Institute *[virtual]*	*Cambridge, MA; Feb. 2026*
† Five College Astronomy Department's Colloquium	*Amherst, MA; Feb. 2026*
† North Houston Astronomy Club (outreach) *[virtual]*	*Humble, TX; Aug. 2025*
† Meeting for NIRCam Instrument Science Team	*Tucson, AZ; Mar. 2025*
† Meeting for MIRI Instrument Science Team	*Tucson, AZ; Oct. 2024*
† JWST Conference, "Cosmic Dawn Revealed"	*Santa Barbara, CA; Aug. 2024*
† Sun City Oro Valley Astronomy Club (outreach)	*Oro Valley, AZ; Mar. 2024*
AAS Meeting #242 (poster)	*Albuquerque, NM; Jun. 2023*
† Steward's Celebration, "100 Years of Astronomy"	*Tucson, AZ; Apr. 2023*
JWST Conference, "A New Era in Astronomy"	*Cambridge, England; Mar. 2023*
† Canyon del Oro High School (outreach)	*Oro Valley, AZ; Dec. 2022*
IAU Symposium #373 (poster)	*Busan, South Korea; Aug. 2022*
AAS Meeting #237 (poster)	*Virtual; Jan. 2021*
AAS Meeting #235 (poster)	*Honolulu, HI; Jan. 2020*
Princeton University Research Day (poster)	*Princeton, NJ; May 2019*
Stanford University Research Conference (poster)	*Stanford, CA; Apr. 2019*
Meeting for Drexel University's Quasar Day	*Philadelphia, PA; Feb. 2019*

MISCELLANEOUS

Programming Languages – Python; IDL; Java; JavaScript; HTML/CSS

Packages, Software, & Tools – Unix; LaTeX; `Prospector`; `BAGPIPES`; `Cloudy`; JWST's Pipeline, Exposure Time Calculator (ETC), and Astronomer's Proposal Tool (APT)

Observational Experience – JWST/MIRI (using the Low Resolution Spectroscopy, or LRS, observing mode); JWST/NIRCam (using the Wide Field Slitless Spectroscopy, or WFSS, observing mode); Keck/MOSFIRE; Magellan/IMACS; Magellan/LDSS3

Soft Skills – academic research; problem solving; public speaking; technical writing

REFERENCES

Prof. Joel Leja	The Pennsylvania State University	joel.leja@psu.edu
Prof. Kevin Hainline	The University of Arizona	kevinhainline@arizona.edu
Prof. Daniel Eisenstein	Harvard University	deisenstein@cfa.harvard.edu
Prof. Jenny Greene	Princeton University	jgreene@astro.princeton.edu
Prof. Marcia Rieke	The University of Arizona	mrieke@gmail.com

PUBLICATIONS (CONTINUED)

Link to SciX Public Library *Last Updated: 3 November 2026*

Summary of All Publications number: 104; citations: 10,595; h-index: 46
Summary of Refereed Publications number: 73; citations: 10,176; h-index: 44

First-Author

[8] **J. M. Helton**, J. Witstok, M. Laska, et al., *Discovery of the Lyman-α Forest in a Star-Forming Galaxy at $z \approx 7.57$: Implications for the Timing and Topology of Cosmic Reionization*, 2026, arXiv:2609.38080

[7] **J. M. Helton**, J. E. Morrison, K. N. Hainline, et al., *Ionizing Photon Production Efficiencies and Chemical Abundances at Cosmic Dawn Revealed by Ultra-Deep Rest-Frame Optical Spectroscopy of JADES-GS-z14-0*, 2026, ApJL, 1004, L30

[6] **J. M. Helton**, S. Alberts, G. H. Rieke, et al., *The Stellar Populations and Rest-Frame Colors of Star-Forming Galaxies at $z \approx 8$: Exploring the Impact of Filter Choice and Star Formation History Assumption with JADES*, 2026, ApJ, 1006, 43

[5] **J. M. Helton**, G. H. Rieke, S. Alberts, et al., *Photometric detection at 7.7 μm of a galaxy beyond redshift 14 with JWST/MIRI*, 2025, Nature Astronomy, 9, 729

[4] **J. M. Helton**, F. Sun, C. Woodrum, et al., *Identification of High-Redshift Galaxy Overdensities in GOODS-N and GOODS-S*, 2024, ApJ, 974, 41

[3] **J. M. Helton**, F. Sun, C. Woodrum, et al., *The JWST Advanced Deep Extragalactic Survey: Discovery of an Extreme Galaxy Overdensity at $z = 5.4$ with JWST/NIRCam in GOODS-S*, 2024, ApJ, 962, 124

[2] **J. M. Helton**, A. L. Strom, J. E. Greene, et al., *The nebular properties of star-forming galaxies at intermediate redshift from the Large Early Galaxy Astrophysics Census*, 2022, ApJ, 934, 81

[1] **J. M. Helton**, S. D. Johnson, J. E. Greene, et al., *Discovery and origins of giant optical nebulae surrounding quasar PKS0454−22*, 2021, MNRAS, 505, 5497

Second-Author

[5] Y. Fudamoto, **J. M. Helton**, X. Lin, et al., *SAPPHIRES: A Galaxy Over-Density in the Heart of Cosmic Reionization at $z = 8.47$*, 2026, ApJ, 1008, 217

[4] K. N. Hainline, **J. M. Helton**, B. E. Miles, et al., *JADES: An Abundance of Ultra-Distant T- and Y-Dwarfs in Deep Extragalactic Data*, 2026, ApJ, 1004, 223

[3] F. D'Eugenio, **J. M. Helton**, K. Hainline, et al., *JADES and SAPPHIRES: galaxy metamorphosis amid a huge, luminous emission-line region*, 2025, MNRAS, 542, 960

[2] K. N. Hainline, **J. M. Helton**, B. D. Johnson, et al., *Brown Dwarf Candidates in the JADES and CEERS Extragalactic Surveys*, 2024, ApJ, 964, 66

[1] F. Sun, **J. M. Helton**, E. Egami, et al., *JADES: Resolving the Stellar Component and Filamentary Overdense Environment of Hubble Space Telescope (HST)-dark Submillimeter Galaxy HDF850.1 at $z = 5.18$*, 2024, ApJ, 961, 69

Co-Author with Major Contributions

[35] O. Curtis, N. J. Cleri, **J. M. Helton**, et al., *A Spectral Framework for Testing the Quasi-Star Hypothesis in Little Red Dots I: Weighing LRDs by Their Super-Eddington Luminosity Ratios — No Signs of Overmassive Black Holes*, 2026, arXiv:2609.09265

[34] K. N. Hainline, S. A. Beiler, et al., including **J. M. Helton**, *JWST/NIRSpec Spectra for Three Ultracool Brown Dwarfs Detected in Extragalactic Surveys: Further Evidence for Phosphine Absorption*, 2026, arXiv:2608.14786

[33] R. Laishram, Y. Koyama, et al., including **J. M. Helton**, *A Compact Proto-group at $z \sim 5$: A Massive Galaxy Caught in Formation*, 2026, ApJ, 1007, L35

[32] Z. Wu, D. J. Eisenstein, et al., including **J. M. Helton**, *JADES: A Prominent Galaxy Overdensity Candidate within the First 500 Myr*, 2026, ApJ, 1006, 18

[31] K. N. Hainline, D. J. Eisenstein, et al., including **J. M. Helton**, *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometrically Selected Galaxy Candidates at $z > 8$*, 2026, ApJ, 1004, 161

[30] F. D'Eugenio, E. J. Nelson, et al., including **J. M. Helton**, *JADES Dark Horse: demonstrating high-multiplex observations with JWST/NIRSpec dense-shutter spectroscopy in the JADES Origins Field*, 2026, MNRAS, 549, stag824

[29] N. J. Cleri, Z. J. Lewis, et al., including **J. M. Helton**, *RUBIES: The Evolution of the Ionization Parameter from $0 < z < 9$*, 2026, arXiv:2605.30410

[28] J. Witstok, S. Carniani, et al., including **J. M. Helton**, *An OASIS of Lyman-α within a Neutral Intergalactic Desert: Reaffirmed Line and Blue Continuum Reveal Efficient Ionising Agents at $z = 13$*, 2026, arXiv:2603.18775

[27] T. Shields, M. J. Rieke, et al., including **J. M. Helton**, *JADES: Low Surface Brightness Galaxies at $0.4 < z < 0.8$ in GOODS-S*, 2026, MNRAS, 546, stag202

[26] X. Lin, E. Egami, et al., including **J. M. Helton**, *The Luminosity Function and Clustering of Hα Emitting Galaxies at $z \approx 4 - 6$ from a Complete NIRCam Grism Redshift Survey*, 2026, ApJ, 997, 207

[25] F. Sun, D. J. Eisenstein, et al., including **J. M. Helton**, *JADES: Discovery of Large Reservoirs of Small Dust Grains in the Circumgalactic Medium of Massive Galaxies at $z \sim 3.5$ through Deep JWST/NIRCam Imaging and Grism Spectroscopy*, 2026, arXiv:2601.15961

[24] S. Alberts, D. J. Eisenstein, et al., including **J. M. Helton**, *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: MIRI Coordinated Parallels in GOODS-S and GOODS-N*, 2026, arXiv:2601.15955

[23] X. Lin, X. Fan, et al., including **J. M. Helton**, *The Large-scale Environments of Low-luminosity AGNs at $3.9 < z < 6.0$ and Implications for Their Host Dark Matter Halos from a Complete NIRCam Grism Redshift Survey*, 2026, ApJ, 997, 61

[22] Y. Zhu, S. Alberts, et al., including **J. M. Helton**, *SMILES: Potentially Higher Ionizing Photon Production Efficiency in Overdense Regions*, 2025, ApJ, 986, 18

[21] J. Witstok, P. Jakobsen, et al., including **J. M. Helton**, *Witnessing the onset of reionization through Lyman-α emission at redshift 13*, 2025, Nature, 639, 897

[20] F. Sun, Y. Fudamoto, et al., including **J. M. Helton**, *Slitless Areal Pure-Parallel HIgh-Redshift Emission Survey (SAPPHIRES): Early Data Release of Deep JWST/NIRCam Images and Spectra in MACS J0416 Parallel Field*, 2025, arXiv:2503.15587

[19] K. N. Hainline, R. Maiolino, et al., including **J. M. Helton**, *An Investigation into the Selection and Colors of Little Red Dots and Active Galactic Nuclei*, 2025, ApJ, 979, 138

[18] J. Witstok, R. Maiolino, et al., including **J. M. Helton**, *JADES: Primaeval Lyman-α emitting galaxies reveal early sites of reionization out to redshift $z \sim 9$*, 2025, MNRAS, 536, 27

[17] S. Alberts, J. Lyu, et al., including **J. M. Helton**, *SMILES Initial Data Release: Unveiling the Obscured Universe with MIRI Multiband Imaging*, 2024, ApJ, 976, 224

[16] K. N. Hainline, F. D'Eugenio, et al., including **J. M. Helton**, *Searching for Emission Lines at $z > 11$: The Role of Damped Lyα and Hints About the Escape of Ionizing Photons*, 2024, ApJ, 976, 160

[15] K. N. Hainline, F. D'Eugenio, et al., including **J. M. Helton**, *JADES: Spectroscopic Confirmation and Proper Motion for a T-Dwarf at 2 kpc*, 2024, ApJ, 975, 31

[14] S. Alberts, C. C. Williams, **J. M. Helton**, et al., *To high redshift and low mass: exploring the emergence of quenched galaxies and their environments at $3 < z < 6$ in the ultra-deep JADES MIRI F770W parallel*, 2024, ApJ, 975, 85

[13] S. Carniani, K. Hainline, et al., including **J. M. Helton**, *Spectroscopic confirmation of two luminous galaxies at a redshift of 14*, 2024, Nature, 633, 318

[12] Z. Ji, C. C. Williams, et al., including **J. M. Helton**, *Extended hot dust emission around the earliest massive quiescent galaxy*, 2024, arXiv:2409.17233

[11] S. Lim, S. Tacchella, et al., including **J. M. Helton**, *The FLAMINGO simulation view of cluster progenitors observed in the epoch of reionization with JWST*, 2024, MNRAS, 532, 4551

[10] B. Robertson, B. D. Johnson, et al., including **J. M. Helton**, *Earliest Galaxies in the JADES Origins Field: Luminosity Function and Cosmic Star-Formation Rate Density 300 Myr after the Big Bang*, 2024, ApJ, 970, 31

[9] K. N. Hainline, B. D. Johnson, et al., including **J. M. Helton**, *The Cosmos in its Infancy: JADES Galaxy Candidates at $z > 8$ in GOODS-S and GOODS-N*, 2024, ApJ, 964, 71

[8] Y. Sun, G.-H. Lee, et al., including **J. M. Helton**, *Evolution of gas flows along the starburst to post-starburst to quiescent galaxy sequence*, 2024, MNRAS, 528, 5783

[7] J. Witstok, R. Smit, et al., including **J. M. Helton**, *Inside the bubble: exploring the environments of reionization-era Lyman-α emitting galaxies with JADES and FRESCO*, 2024, A&A, 682, A40

[6] Z. Li, Z. Cai, et al., including **J. M. Helton**, *MAGNIF: A Tentative Lensed Rotating Disk at $z = 8.34$ detected by JWST NIRCam WFSS with Dynamical Forward Modeling*, 2023, arXiv:2310.09327

[5] L. Sandles, F. D'Eugenio, **J. M. Helton**, et al., *JADES: deep spectroscopy of a low-mass galaxy at redshift 2.3 quenched by environment*, 2023, arXiv:2307.08633

[4] S. Tacchella, D. J. Eisenstein, et al., including **J. M. Helton**, *JADES Imaging of GN-z11: Revealing the Morphology and Environment of a Luminous Galaxy 430 Myr After the Big Bang*, 2023, ApJ, 952, 74

[3] E. Curtis-Lake, S. Carniani, et al., including **J. M. Helton**, *Spectroscopic confirmation of four metal-poor galaxies at $z = 10.3 - 13.2$*, 2023, Nature Astronomy, 7, 622

[2] B. E. Robertson, S. Tacchella, et al., including **J. M. Helton**, *Identification and properties of intense star-forming galaxies at $z > 10$*, 2023, Nature Astronomy, 7, 611

[1] S. Aiola, E. Calabrese, et al., including **J. M. Helton**, *The Atacama Cosmology Telescope: DR4 Maps and Cosmological Parameters*, 2020, JCAP, 12, 047

Co-Author with Minor Contributions

[56] W. M. Baker, T. J. Looser, et al., including **J. M. Helton**, *OASIS III: Bursty Star Formation Drives Observed Galaxy Sizes at $z > 4$*, 2026, arXiv:2610.41491

[55] T. J. Looser, F. D'Eugenio, et al., including **J. M. Helton**, *OASIS II: The Ionized Gas Traces Feedback-Driven Declines into the Quiet Side of Bursty Star Formation at $z > 4$*, 2026, arXiv:2610.41490

[54] T. J. Looser, F. D'Eugenio, et al., including **J. M. Helton**, *OASIS I: Spectroscopy of Mini-Quenched and Lulling Galaxies Constrains the Quiet Side of Bursty Star Formation at $z > 4$*, 2026, arXiv:2610.41489

[53] P. Rinaldi, J. Álvarez-Márquez, et al., including **J. M. Helton**, *PRISMS/U13151. A Glimpse into the Earliest Phases of Galaxy Formation through an Ultra-Faint System $\approx$ 500 Myr after the Big Bang*, 2026, arXiv:2610.26560

[52] F. Sun, J. Álvarez-Márquez, et al., including **J. M. Helton**, *An Archival Calibration of JWST/MIRI Prism Wide-Field Slitless Spectroscopy: Methodology, Perfor-*

mance, and a Mid-Infrared Spectral Atlas of Galaxies at $z = 0 - 4$ in the GOODS-N/S Fields, 2026, arXiv:2609.20935

[51] Z. Wu, D. J. Eisenstein, et al., including **J. M. Helton**, *JADES: Dynamical Measurements of Dark Matter Halo Masses at $z \approx 7$ Using Galaxy Pairs*, 2026, arXiv:2609.17756

[50] B. Pérez-Díaz, M. Castellano, et al., including **J. M. Helton**, *PRISMS. GHZ1. A standard tale of galaxy evolution with atypical ionizing conditions at $z = 9.878$*, 2026, arXiv:2609.14574

[49] J. A. A. Trussler, D. J. Eisenstein, et al., including **J. M. Helton**, *Out of oxygen: extremely metal-poor galaxy candidates identified at $2.5 < z < 6.5$ with deep JADES medium-band imaging*, 2026, MNRAS, 550, stag1360

[48] Y. Zhu, X. Fan, et al., including **J. M. Helton**, *Low Lyα Visibility in Galaxy Overdensities: Reionization Topology and Neutral-Fraction Ceilings from DIVER over $4.8 < z < 11$*, 2026, arXiv:2608.19311

[47] O. Curtis, A. J. Rowland, et al., including **J. M. Helton**, *The $\hat{G}$ Infrared Search for Extraterrestrial Civilizations with Large Energy Supplies. V. When Galaxies Glow with Industry*, 2026, arXiv:2608.12458

[46] C. Witten, J. S. Bennett, et al., including **J. M. Helton**, *Too many protoclusters? Reconciling the overabundance of cluster progenitors within the first billion years of the Universe*, 2026, arXiv:2605.28930

[45] Q. Duan, S. Tacchella, et al., including **J. M. Helton**, *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: stellar population catalogue for galaxies in GOODS-N and GOODS-S*, 2026, arXiv:2605.21599

[44] S. Carniani, P. Jakobsen, et al., including **J. M. Helton**, *Intense and Extended CIII] Emission Suggests a Strong Outflow in JADES-GS-z14-0*, 2026, arXiv:2604.11899

[43] P. Rinaldi, K. N. Hainline, et al., including **J. M. Helton**, *The Way We Tally Becomes the Tale: The Impact of Selection Strategies on the Inferred Evolution of Little Red Dots Across Cosmic Time*, 2026, arXiv:2604.07138

[42] S. Alberts, G. H. Rieke, et al., including **J. M. Helton**, *Calibrating Photometric Mid-Infrared Star Formation Rates for JWST*, 2026, arXiv:2604.01326

[41] D. J. Eisenstein, C. Willott, et al., including **J. M. Helton**, *Overview of the JWST Advanced Deep Extragalactic Survey (JADES)*, 2026, ApJS, 283, 6

[40] R. Maiolino, H. Übler, et al., including **J. M. Helton**, *The Search for Population III: Confirmation of a HeII Emitter with No Metal Lines at $z = 10.6$*, 2026, arXiv:2603.20362

[39] Z. Ji, C. C. Williams, et al., including **J. M. Helton**, *JADES: Rest-frame UV-to-NIR Size Evolution of Massive Quiescent Galaxies from Redshift $z = 5$ to $z = 0.5$*, 2026, ApJ, 998, 239

[38] Y. Zhu, N. Bonaventura, et al., including **J. M. Helton**, *SMILES Data Release II: Probing Galaxy Evolution during Cosmic Noon and Beyond with NIRSpec Medium-Resolution Spectra*, 2026, ApJ, 997, 301

[37] J. Zhang, E. Egami, et al., including **J. M. Helton**, *Abundant Population of Broad Hα Emitters in the GOODS-N Field Revealed by CONGRESS, FRESCO, and JADES*, 2026, ApJ, 997, 250

[36] B. Wang, J. Leja, et al., including **J. M. Helton**, *Water Absorption Confirms Cool Atmospheres in Two Little Red Dots*, 2026, arXiv:2602.06024

[35] B. E. Robertson, B. D. Johnson, et al., including **J. M. Helton**, *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometric Catalog*, 2026, arXiv:2601.15956

[34] B. D. Johnson, B. E. Robertson, et al., including **J. M. Helton**, *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: NIRCam Imaging in GOODS-S and GOODS-N*, 2026, arXiv:2601.15954

[33] J. Witstok, R. Smit, et al., including **J. M. Helton**, *On the origins of oxygen: ALMA and JWST characterize the multi-phase, metal-enriched, star-bursting medium within a 'normal' $z > 11$ galaxy*, 2026, OJAp, 9, 55261

[32] D. J. Eisenstein, B. D. Johnson, et al., including **J. M. Helton**, *The JADES Origins Field: A New JWST Deep Field in the JADES Second NIRCam Data Release*, 2025, ApJS, 281, 50

[31] Y. Zhu, E. Egami, et al., including **J. M. Helton**, *Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$*, 2025, ApJ, 995, L5

[30] C. Simmonds, S. Tacchella, et al., including **J. M. Helton**, *Bursting at the seams: the star-forming main sequence and its scatter at $z = 3 - 9$ using NIRCam photometry from JADES*, 2025, MNRAS, 544, 4551

[29] A. L. Danhaive, S. Tacchella, et al., including **J. M. Helton**, *The dawn of disks: unveiling the turbulent ionized gas kinematics of the galaxy population at $z \sim 4 - 6$ with JWST/NIRCam grism spectroscopy*, 2025, MNRAS, 543, 3249

[28] Z. Wu, D. J. Eisenstein, et al., including **J. M. Helton**, *JADES-GS-z14-1: A Compact, Faint Galaxy at $z \approx 14$ with Weak Metal Lines from Extremely Deep JWST MIRI, NIRCam, and NIRSpec Observations*, 2025, ApJ, 992, 212

[27] P. Rinaldi, N. Bonaventura, et al., including **J. M. Helton**, *Not Just a Dot: The Complex UV Morphology and Underlying Properties of Little Red Dots*, 2025, ApJ, 992, 71

[26] L. Whitler, D. P. Stark, et al., including **J. M. Helton**, *The $z \geq 9$ Galaxy UV Luminosity Function from the JWST Advanced Deep Extragalactic Survey: Insights Into Early Galaxy Evolution and Reionization*, 2025, ApJ, 992, 63

[25] C. Woodrum, I. Shivaei, et al., including **J. M. Helton**, *JADES: The Star Formation and Dust Attenuation Properties of Galaxies at $3 < z < 7$*, 2025, arXiv:2510.00235

[24] K. Ormerod, J. Witstok, et al., including **J. M. Helton**, *Detection of the 2175 Å UV Bump at $z > 7$: evidence for rapid dust evolution in a merging reionization-era galaxy*, 2025, MNRAS, 542, 1136

[23] Z. Ji, S. Alberts, et al., including **J. M. Helton**, *The Importance of Dust Distribution in Ionizing-photon Escape: NIRCam and MIRI Imaging of a Lyman Continuum-emitting Galaxy at $z \sim 3.8$*, 2025, ApJ, 988, L69

[22] S. Tacchella, W. McClymont, et al., including **J. M. Helton**, *Resolving the nature and putative nebular emission of GS9422: an obscured AGN without exotic stars*, 2025, MNRAS, 540, 851

[21] Y. Zhu, M. J. Rieke, et al., including **J. M. Helton**, *A Systematic Search for Galaxies with Extended Emission Lines and Potential Outflows in JADES Medium-band Images*, 2025, ApJ, 986, 162

[20] T. Y.-Y. Hsiao, F. Sun, et al., including **J. M. Helton**, *SAPPHIRES: Extremely Metal-Poor Galaxy Candidates with $12 + log\,(O/H) < 7.0$ at $z \sim 5 - 7$ from Deep JWST/NIRCam Grism Observations*, 2025, arXiv:2505.03873

[19] S. Carniani, F. D'Eugenio, et al., including **J. M. Helton**, *The eventful life of a luminous galaxy at $z = 14$: metal enrichment, feedback, and low gas fraction?*, 2025, A&A, 696, A87

[18] F. D'Eugenio, A. J. Cameron, et al., including **J. M. Helton**, *JADES Data Release 3: NIRSpec/Microshutter Assembly Spectroscopy for 4000 Galaxies in the GOODS Fields*, 2025, ApJS, 277, 4

[17] F. D'Eugenio, R. Maiolino, et al., including **J. M. Helton**, *JWST/NIRSpec WIDE survey: a $z = 4.6$ low-mass star-forming galaxy hosting a jet-driven shock with low ionization and solar metallicity*, 2025, MNRAS, 536, 51

[16] Y. Sun, J. Lyu, et al., including **J. M. Helton**, *No Evidence for a Significant Evolution of $M_\bullet - M_*$ Relation in Massive Galaxies up to $z \sim 4$*, 2025, ApJ, 978, 98

[15] W. M. Baker, S. Tacchella, et al., including **J. M. Helton**, *A core in a star-forming disc as evidence of inside-out growth in the early Universe*, 2025, Nature Astronomy, 9, 141

[14] Z. Ji, C. C. Williams, et al., including **J. M. Helton**, *JADES+JEMS: A Detailed Look at the Buildup of Central Stellar Cores and Suppression of Star Formation in Galaxies at Redshifts $3 < z < 4.5$*, 2024, ApJ, 974, 135

[13] A. J. Bunker, A. J. Cameron, et al., including **J. M. Helton**, *JADES NIRSpec initial data release for the Hubble Ultra Deep Field: redshifts and line fluxes of distant galaxies from the deepest JWST Cycle 1 NIRSpec multi-object spectroscopy*, 2024, A&A, 690, A288

[12] C. Woodrum, M. J. Rieke, et al., including **J. M. Helton**, *JADES: Using NIRCam photometry to investigate the dependence of stellar mass inferences on the IMF in the early Universe*, 2024, PNAS, 121, id.e2317375121

[11] R. Endsley, D. P. Stark, et al., including **J. M. Helton**, *The star-forming and ionizing properties of dwarf $z \sim 6-9$ galaxies in JADES: insights on bursty star formation and ionized bubble growth*, 2024, MNRAS, 533, 1111

[10] C. C. Williams, S. Alberts, et al., including **J. M. Helton**, *The Galaxies Missed by Hubble and ALMA: The Contribution of Extremely Red Galaxies to the Cosmic Census at $3 < z < 8$*, 2024, ApJ, 968, 34

[9] A. Saxena, A. J. Bunker, et al., including **J. M. Helton**, *JADES: The production and escape of ionizing photons from faint Lyman-alpha emitters in the epoch of reionization*, 2024, A&A, 684, A84

[8] M. Curti, R. Maiolino, et al., including **J. M. Helton**, *JADES: Insights into the low-mass end of the mass-metallicity-SFR relation at $3 < z < 10$ from deep JWST/NIRSpec spectroscopy*, 2024, A&A, 684, A75

[7] R. Maiolino, J. Scholtz, et al., including **J. M. Helton**, *A small and vigorous black hole in the early Universe*, 2024, Nature, 627, 59

[6] M. J. Rieke, B. Robertson, et al., including **J. M. Helton**, *JADES Initial Data Release for the Hubble Ultra Deep Field: Revealing the Faint Infrared Sky with Deep JWST NIRCam Imaging*, 2023, ApJS, 269, 16

[5] C. C. Williams, S. Tacchella, et al., including **J. M. Helton**, *JEMS: A Deep Medium-band Imaging Survey in the Hubble Ultra Deep Field with JWST NIRCam and NIRISS*, 2023, ApJS, 268, 64

[4] A. J. Cameron, A. Saxena, et al., including **J. M. Helton**, *JADES: Probing interstellar medium conditions at $z \approx 5.5-9.5$ with ultra-deep JWST/NIRSpec spectroscopy*, 2023, A&A, 677, A115

[3] A. J. Bunker, A. Saxena, et al., including **J. M. Helton**, *JADES NIRSpec Spectroscopy of GN-z11: Lyman-α emission and possible enhanced nitrogen abundance in a $z = 10.60$ luminous galaxy*, 2023, A&A, 677, A88

[2] S. Tacchella, B. D. Johnson, et al., including **J. M. Helton**, *JWST NIRCam+NIRSpec: interstellar medium and stellar populations of young galaxies with rising star formation and evolving gas reservoirs*, 2023, MNRAS, 522, 6236

[1] B. E. Robertson, S. Tacchella, et al., including **J. M. Helton**, *Morpheus Reveals Distant Disk Galaxy Morphologies with JWST: The First AI/ML Analysis of JWST Images*, 2023, ApJ, 942, L42